S. P. Mandali's

RAMANARAIN RUIA AUTONOMOUS COLLEGE
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Department of Computer Science & IT

Project Report On

FoodRescue Hub

Submitted By

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SUBMITTED IN PARTIAL FULFILLMENT OF

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S. P. Mandali's

RAMNARAIN RUIA AUTONOMOUS COLLEGE

Matunga

Department of
Computer Science & Information Technology

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Abstract

1.1 Problem Statement

Edible food is discarded in massive quantities daily by hospitality establishments, commercial food vendors, event hosts, and institutional cafeterias, even as local communities struggle with critical nutritional deficits. Traditional donation workflows rely almost exclusively on ad-hoc phone calls, informal instant messaging groups, and un-synchronized spreadsheets. This creates high communication latency, complete lack of operational tracking, and inevitable spoilage of perishable food before logistical collection can be arranged. Small-to-medium Non-Governmental Organizations (NGOs) lack unified digital visibility into incoming donation availability, leading to severe logistical bottlenecks and uncoordinated redistribution.

1.2 Abstract Summary

The **Food Donation and Expiry Management System** is a centralized, role-based web architecture designed to automate, synchronize, and expedite the end-to-end lifecycle of surplus food distribution between donors, verified NGOs, and volunteer logistics networks.

Unlike generalized inventory systems, this platform implements **active, rule-based shelf-life decay monitoring**, computing real-time expiry thresholds from precise preparation timestamps and item perishability classifications. The platform incorporates:

1. **Dynamic Alert Engine:** Classifies donations into active, near-expiry (urgent thresholds < 2 hours), and auto-expired states to eliminate human dispatch errors.
2. **Transparent Role-Based Coordination:** Synchronizes Donor donation requests, NGO geographic claims, and Volunteer route assignments.
3. **Auditable Metrics Engine:** Tracks delivery timestamps, beneficiary metrics, and aggregate food waste avoided.

The platform made donation acceptance faster and ensured that no donations were missed or left unrecorded.

Introduction

2.1 Aims and Objectives of the Research

- **Aim:** To design, build, and empirically evaluate a centralized, real-time web platform that minimizes surplus food wastage by streamlining communication, tracking perishable lifespans, and automating logistical coordination between donors and charitable distributors.
- **Objectives:**
    1. Analyze limitations and structural failures in conventional, unorganized donation systems.
    2. Synthesize operational requirements across donor classes, humanitarian NGOs, and volunteer networks.
    3. Formulate a relational data architecture supporting real-time status transitions and expiry tracking.
    4. Develop automated rule-based shelf-life monitoring algorithms with urgent notifications and automatic expiry invalidation.
    5. Implement and deploy a scalable, multi-role web application using a modern full-stack ecosystem.
    6. Conduct quantitative and qualitative benchmark evaluations measuring time-to-acceptance, data precision, and operational usability compared to legacy workflows.

2.2 Background and Motivation of Research Domain

According to the United Nations Environment Programme (UNEP) Food Waste Index and FAO figures, roughly one-third of all food produced globally is lost or wasted, directly conflicting with Sustainable Development Goals **SDG 2 (Zero Hunger)** and **SDG 12 (Responsible Consumption and Production)**. Paradoxically, urban hunger persists in parallel with commercial food surplus. The core failure is logistical and informational: surplus food is time-sensitive and highly perishable. Without instantaneous data dissemination and structured distribution management, edible surplus becomes organic waste within hours.

2.3 Topic / Area / Domain for Investigation

- **Domain:** Information Systems, Applied Web Engineering, Humanitarian Logistics, and Waste Management Informatics.
- **Research Focus:** Real-time state machine lifecycle management, time-sensitive inventory dispatching, and relational transparency in charitable supply chains.

2.4 Research Problem Statement

How can a centralized, time-indexed web platform resolve communication bottlenecks, eliminate data integrity errors, and ensure the rapid, safe redistribution of perishable food donations before safety thresholds expire?

Significance, Impact, and Future Relevance

- **Social Impact:** Enables direct, structured diversion of wholesome food to underprivileged communities and shelters.
- **Environmental Impact:** Prevents methane emissions generated by organic waste decomposing in municipal landfills.
- **Future Relevance:** Provides a modular foundation for integrating IoT sensor nodes (cold-chain temperature loggers), dynamic routing heuristics for fleet dispatch, and predictive machine-learning engines for demand forecasting.
- **In-Scope:** Role-based access control (Admin, Donor, NGO, Volunteer); automated expiry alerts; live status tracking from dispatch to handover; database logging; administrative compliance audits.
- **Limitations:** The initial implementation relies on donor-declared preparation and expiry timestamps without automated chemical sensor verification; route planning assumes local city navigation without built-in automated fleet multi-stop optimization.
- **Technical Feasibility:** The architecture utilizes industry-standard open-source web stacks (e.g., Python Flask/Node.js backend, relational PostgreSQL/MySQL, responsive HTML5/Bootstrap frontend) with negligible resource overhead.
- **Operational Feasibility:** Minimal operational barrier for end users; intuitive interfaces allow non-technical volunteers and donors to post and claim donations in fewer than three clicks.
- **Economic Feasibility:** Built entirely on open-source frameworks, permitting community deployment with zero recurring proprietary licensing charges.

2.5 Technology Stack Justification

- **Presentation Tier:** React 19 Single Page Application (SPA) driven by Vite 6 and styled using Tailwind CSS v4 with Lucide Icons. This stack eliminates full-page browser reloads, delivers sub-millisecond DOM reconciliations, and dynamically computes visual urgency indicators across mobile and desktop devices.
- **Application Tier:** Node.js (LTS v24) and Express.js REST API framework. Express handles stateless HTTP request pipelines, integrates Bearer JSON Web Token (JWT) authentication, and executes an autonomous 60-second background daemon to evaluate real-time expiry decay curves.
- **Data & Persistence Tier:** sql.js (SQLite 3 compiled to WebAssembly / WASM). This provides an embedded, ACID-compliant relational SQL engine executing in-memory with periodic binary disk persistence (food_donation.db). It eliminates external database server overheads and connection latency during local deployment and evaluation while preserving full relational foreign-key integrity.

Literature Review

3.1 Overview of Existing Research Related to Topic

Computational interventions in food salvage range from simple noticeboard applications to complex GIS-enabled logistics platforms. While commercial supply chain software emphasizes non-perishable inventory management, charitable redistribution requires hyper-localized, ultra-fast matching due to stringent perishable timelines. Academic research demonstrates that traditional methods (spreadsheets, messaging apps) break down rapidly once donation frequencies cross minimal thresholds.

3.2 Review of Key Literature (Summary of Five Foundational Papers)

1. **Muniandy & Arbaiy (2022)** — *Development of Information Management System for Food Donation Distribution*:

- Investigated structural failures within NGOs relying on Excel spreadsheets.
- Proved that manual logging causes file version corruption, lost historical records, and severe delays in recipient dispatch.
- Established the necessity of structured relational data layers and authenticated role-based controls.

2. **Swetha & Pavana (2026)** — *Smart Food Waste Management System with Real-Time NGO Notification and Donation Tracking*:

- Implemented a modern reactive web architecture connecting food businesses directly with local charity organizations.
- Emphasized that recording preparation and expiry times directly enables automated alerts, dramatically reducing perishable transit failures.

3. **Shaikh, Shaikh, & Rayeen (2026)** — *Leftover Food Donation Management System*:

- Explored operational barriers in community-level surplus reallocation.
- Highlighted that lack of designated volunteer-assignment modules creates dispatch bottlenecks where claimed food sits uncollected.

4. **Shivani & Usha (2026)** — *A Web-Based Food Donation and Redistribution System to Minimize Food Waste and Support NGOS*:

- Addressed the relationship between computational redistribution platforms and UN Sustainable Development Goals (SDG 2 and 12).
- Demonstrated the utility of geographic filtering and automated messaging in lowering time-to-pickup.

5. **Pol et al. (2025)** — *Food Connect+: A Progressive Framework for Efficient Food Donation and NGO Collaboration*:

- Examined iterative human-computer interaction (HCI) improvements on early donation prototypes.
- Showed that user-retention and system utility depend on workflow clarity, reporting transparency, and immediate operational status feedback.

3.3 Comparative Summary Table of Literature

Author(s) & Year	Proposed Architecture	Key Focus / Strength	Identified Limitations / Research Gap
Muniandy & Arbaiy (2022)	Web-based Information System (PHP/MySQL)	Replacement of spreadsheets; role-based recipient management.	Lacked real-time dynamic countdown timers and automated expiry status invalidation.
Swetha & Pavana (2026)	React.js, Supabase, PowerBI	Real-time notifications; analytical dashboarding.	High dependency on third-party cloud services; limited volunteer logistics tracking.
Shaikh et al. (2026)	Multi-tier Web Application	Volunteer coordination and administrative monitoring.	Minimal focus on food safety checklists or empirical evaluation studies.
Shivani & Usha (2026)	Location-aware Web Portal	Spatial filtering and alignment with UN Sustainable Goals.	Absence of strict state-locking to prevent concurrent claim collisions
Pol et al. (2025)	Progressive Iterative Framework	Usability enhancements, workflow optimization.	Focused on prototype iteration without strict quantitative baseline comparative analysis.

3.4 Gap Identification

Existing systems either focus purely on static record-keeping or general logistics without enforcing time-critical perishability dynamics. Prior systems frequently treat "Expiry Time" as a passive text label rather than an **active state driver**. A system is needed that actively computes expiry differentials, provides automated threshold alerts, blocks expired collections automatically, and coordinates volunteer pickup within a single lightweight, accessible platform.

Research Methodology

4.1 Description of Algorithms, Logic, and Procedures

1. Dynamic Expiry Evaluation & State Mutation Algorithm

The platform continuously tracks donation validity via an automated calculation engine:

- Let $T_{current}$ be the current system time.
- Let T_{exp} be the user-declared food expiration timestamp.
- Let $\Delta T = T_{exp} - T_{current}$.

Algorithm: EvaluateDonationStatus(DonationID)

1. Fetch record for DonationID from Database
2. If record.status in ['Delivered', 'Cancelled'] -> Terminate
3. Calculate DeltaT = record.expiry_datetime - System.CurrentTime()
4. If DeltaT <= 0:

    Update record.status = 'Expired'

    Emit StatusChangeAlert(DonationID, 'Expired')
5. Else If DeltaT <= 2 Hours:

    Mark record.alert_priority = 'URGENT'

    Emit Notification(NGO_Pool, 'Item nearing expiration!')
6. Else:

    Mark record.alert_priority = 'NORMAL'
6. Persist record changes

2. Strict State-Machine Progression Logic

To preserve transactional integrity, state changes follow a deterministic, irreversible pipeline:

$$\text{Posted} \longrightarrow \text{Accepted} \longrightarrow \text{Volunteer Assigned} \longrightarrow \text{Picked Up} \longrightarrow \text{Delivered}$$

- At any stage prior to "Picked Up", if $\Delta T \leq 0$, the state transitions automatically to **Expired**.

4.2 Data Collection and Requirements Gathering

- **Functional Requirements:**
    1. *Donor Module:* Secure authentication, structured donation posting (food type, quantity, preparation timestamp, expiry timestamp, pickup address, notes).
    2. *NGO Module:* Search/filter available donations by proximity, category, and urgency; atomic donation claiming; volunteer dispatch.
    3. *Volunteer Module:* View assigned pickup missions, submit status updates ("Picked Up", "Delivered").
    4. *Admin Module:* NGO credential vetting, site-wide audit logs, aggregated statistical distribution reporting.
- **Non-Functional Requirements:**
    1. High concurrency locking to prevent duplicate claims.
    2. Sub-second query responses on available listings.
    3. Responsive UI performance on mobile and desktop viewports.

4.3 Database Schema and Relational Design

```
================================================================

                        DATABASE TABLES

================================================================

1. USERS: Authentication and RBAC Core

CREATE TABLE users (
   id INTEGER PRIMARY KEY AUTOINCREMENT,
   name TEXT NOT NULL,
   email TEXT UNIQUE NOT NULL,
   password_hash TEXT NOT NULL,
   role TEXT CHECK(role IN ('donor', 'ngo', 'volunteer', 'admin')) NOT NULL,
   phone TEXT,
   created_at DATETIME DEFAULT CURRENT_TIMESTAMP
);
```

2. NGOS: Profile & Administrative Compliance

```
CREATE TABLE ngos (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    user_id INTEGER UNIQUE NOT NULL,
    ngo_name TEXT NOT NULL,
    registration_number TEXT,
    address TEXT NOT NULL,
    city TEXT NOT NULL,
    verified INTEGER DEFAULT 0 CHECK(verified IN (0, 1)),
    FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE
);
```

3. DONATIONS: Perishable Inventory Records

```
CREATE TABLE donations (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    donor_id INTEGER NOT NULL,
    food_name TEXT NOT NULL,
    food_type TEXT NOT NULL,
    quantity TEXT NOT NULL,
    preparation_datetime DATETIME NOT NULL,
    expiry_datetime DATETIME NOT NULL,
    pickup_location TEXT NOT NULL,
    image_url TEXT,
    status TEXT DEFAULT 'posted' CHECK(status IN ('posted', 'accepted', 'assigned', 'picked_up', 'delivered', 'expired', 'cancelled')),
    created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (donor_id) REFERENCES users(id) ON DELETE CASCADE
);
```

4. DONATION_ASSIGNMENTS: Logistics Tracking

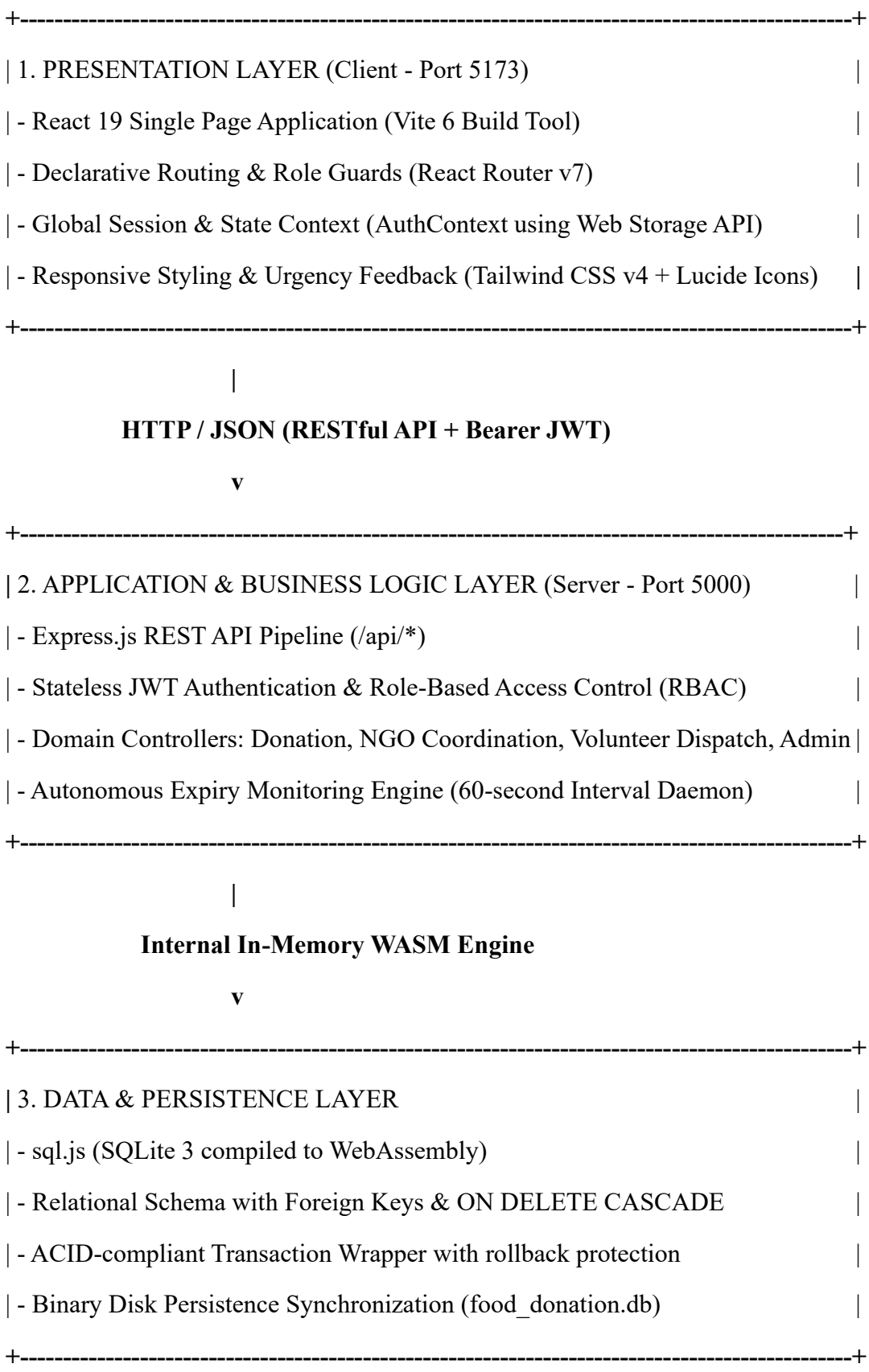
```
CREATE TABLE donation_assignments (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    donation_id INTEGER UNIQUE NOT NULL,
    ngo_id INTEGER NOT NULL,
    volunteer_id INTEGER,
    collection_type TEXT CHECK(collection_type IN ('self_pickup', 'volunteer')) NOT NULL,
    delivery_notes TEXT,
    accepted_at DATETIME DEFAULT CURRENT_TIMESTAMP,
    delivered_at DATETIME,
    beneficiaries_served INTEGER,
    FOREIGN KEY (donation_id) REFERENCES donations(id) ON DELETE CASCADE,
    FOREIGN KEY (ngo_id) REFERENCES ngos(id) ON DELETE CASCADE,
    FOREIGN KEY (volunteer_id) REFERENCES users(id) ON DELETE SET NULL
);
```

5. ACTIVITY_LOGS: Audit Trail

```
CREATE TABLE activity_logs (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    donation_id INTEGER,
    user_id INTEGER,
    action TEXT NOT NULL,
    details TEXT,
    timestamp DATETIME DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (donation_id) REFERENCES donations(id) ON DELETE SET NULL,
    FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE SET NULL);
```

==========

4.4 Layered System Architecture



Experimental Setup

The experimental and development setup was engineered to ensure high performance, rapid compilation, cross-platform reproducibility, and zero dependency on external cloud database servers during local development and academic evaluation.

5.1 Tools and Execution Environment

5.1.1 Hardware Specifications

- Processor (CPU): Intel Core / AMD Ryzen Multi-Core x86_64 architecture (base clock frequency $\geq$ 2.4 GHz).
- Random Access Memory (RAM): 16.0 GB DDR4/DDR5 system memory.
- Secondary Storage: High-speed NVMe Solid State Drive (SSD) with $\geq$ 500 MB allocated for source code, node_modules dependencies, and SQLite database storage.
- Network Interface: Localhost network interface (127.0.0.1 loopback) simulating high-throughput, zero-latency HTTP communication between client and server.

5.1.2 Software Execution Environment

- Operating System: Microsoft Windows 10 / 11 64-bit Edition with PowerShell 7 and Windows Terminal.
- Server Runtime: Node.js LTS (v24.12.0) executing the Google V8 JavaScript engine with native ES2023+ standard support.
- Package Manager: npm (v11.6.2) for package resolution, dependency locking, and automated lifecycle script execution.
- Client Environment: Evergreen WebAssembly-capable web browsers:
    - Google Chrome (Version 120+)
    - Microsoft Edge (Chromium-based, Version 120+)
    - Mozilla Firefox (Version 120+)
- Integrated Development Environment (IDE): Visual Studio Code / Antigravity IDE with Tailwind CSS IntelliSense, ESLint, and SQLite Data Viewer extensions.
- Testing and API Verification Tools: Windows PowerShell (Invoke-RestMethod), curl CLI, and browser developer console for end-to-end network, header, and JWT token validation.

5.2 Architecture and Framework

The platform uses a Three-Tier Decoupled Client-Server Architecture communicating over stateless RESTful HTTP protocols with JSON serialization.

5.2.1 Presentation Layer (Frontend SPA)

- Single Page Application (SPA): Built using React 19 and compiled via Vite 6, eliminating full browser reloads and providing seamless role-specific dashboards.
- Global Authentication Context: React Context (AuthContext) manages user sessions, decodes JWT tokens, and transparently injects Bearer authorization headers into API calls.
- Responsive Design System: Powered by Tailwind CSS v4, dynamically adapting across mobile viewports, tablets, and desktop displays with color-coded urgency states.

5.2.2 Application Layer (Backend API)

- RESTful Micro-Routing: Built on Express.js, separating routing (api.js), controller logic (donationController, ngoController), and background services (expiryService).
- Security & RBAC: Intercepts privileged endpoints with cryptographic JWT token verification, preventing privilege escalation between donors, NGOs, volunteers, and admins.
- Background Expiry Daemon: A recurring daemon runs every 60 seconds, systematically evaluating timestamp differentials and transitioning unclaimed food to 'expired'.

5.2.3 Data Layer (Relational Storage)

- WebAssembly Database Engine: Utilizes sql.js (SQLite compiled to WebAssembly), providing full SQL compliance (joins, constraints, transactions) without requiring native C++ build tools or external server daemons.
- Transactional Integrity: Multi-stage business operations run inside ACID transactions (BEGIN TRANSACTION ... COMMIT) with deferred file writes to guarantee data durability.

5.3 Software Language and Library Specifications

5.3.1 Core Programming Languages

- JavaScript (ECMAScript 2023+): Used across client (ES Modules) and server (CommonJS runtime) for asynchronous non-blocking event-driven processing.
- SQL (ANSI / SQLite Dialect): Used for table definitions, referential integrity constraints, foreign key indexing, and statistical aggregations.
- HTML5 & CSS3: Semantic markup and modern styling primitives (Flexbox, CSS Grid).

5.3.2 Frontend Dependencies (client/package.json)

- react (v19.0.0) & react-dom (v19.0.0): Core framework for declarative, component-driven user interfaces and DOM reconciliation.

- vite (v6.2.0): Build tool and local development server providing instant Hot Module Replacement (HMR).
- react-router-dom (v7.2.0): Declarative client-side routing, navigation state, and role-protected Route Guards.
- tailwindcss (v4.0.9) & @tailwindcss/vite (v4.0.9): Modern utility-first CSS framework for responsive layout rendering.
- lucide-react (v1.16.0): SVG icon suite used for operational indicators, urgency badges, and UI controls.

5.3.3 Backend Dependencies (server/package.json)

- express (v4.21.2): Web framework for RESTful routing, middleware chaining, and request pipeline handling.
- sql.js (v1.12.0): SQLite relational database engine compiled to WebAssembly (WASM), enabling local relational execution with persistent binary file export.
- jsonwebtoken (v9.0.2): Issues, signs, and verifies HMAC-SHA256 authenticated JSON Web Tokens (JWT) adhering to RFC 7519.
- bcryptjs (v2.4.3): Blowfish-based adaptive cryptographic password hashing algorithm (configured with 10 salt rounds).
- cors (v2.8.5): Cross-Origin Resource Sharing middleware enabling secure API calls between client (localhost:5173) and server (localhost:5000).
- dotenv (v16.4.7): Loads system environment variables from .env files into Node's runtime context.
- multer (v1.4.5-lts.1): Middleware handling multipart/form-data for food image uploads.

5.4 Implementation Code Snippets

5.4.1 Dynamic Expiry Algorithm and Background Daemon

```
const db = require('../config/database');

/**
 * Calculates time delta and assigns urgency metadata
 * Mathematical model: Delta_t = t_expiry - t_now
 */
function getUrgencyInfo(expiryDateTimeStr) {
  const now = new Date();
  const expiry = new Date(expiryDateTimeStr);
  const diffMinutes = Math.floor((expiry - now) / (1000 * 60));
```

```
  if (diffMinutes <= 0) {
    return { urgency: 'expired', diffMinutes: 0, badgeColor: 'gray', isUrgent: true, label: 'Expired' };
  } else if (diffMinutes <= 120) {
    // Under 2 Hours: High-urgency threshold
    return { urgency: 'urgent', diffMinutes, badgeColor: 'red', isUrgent: true, label: 'Urgent (< 2 hrs)' };
  } else if (diffMinutes <= 240) {
    // 2 to 4 Hours: Warning threshold
    return { urgency: 'expiring_soon', diffMinutes, badgeColor: 'amber', isUrgent: false, label: 'Expiring
Soon' };
  } else {
    // Greater than 4 Hours: Safe window
    return { urgency: 'fresh', diffMinutes, badgeColor: 'emerald', isUrgent: false, label: 'Fresh' };
  }
}
/**
 * Autonomous background sweep: Flags unrescued food as expired
 */
function checkAndUpdateExpiredDonations() {
  const nowIso = new Date().toISOString();

  const expiredDonations = db.prepare(`
    SELECT id, food_name FROM donations
    WHERE status = 'posted' AND expiry_datetime <= ?
  `).all(nowIso);
  if (expiredDonations.length > 0) {
    const updateStmt = db.prepare(`UPDATE donations SET status = 'expired' WHERE id = ?`);
    const logStmt = db.prepare(`
      INSERT INTO activity_logs (donation_id, user_id, action, details)
      VALUES (?, NULL, 'AUTO_EXPIRED', ?)
    `);
    // Execute atomic batch update
    const autoExpireTx = db.transaction((donations) => {
      for (const item of donations) {
        updateStmt.run(item.id);
```

```
      logStmt.run(item.id, `Donation "${item.food_name}" automatically marked expired due to elapsed time.`);
    }
  });
  autoExpireTx(expiredDonations);
  console.log(`[ExpiryMonitor] Automatically expired ${expiredDonations.length} unclaimed donation(s).`);
 }
}
// Start monitor running every 60 seconds
function startExpiryMonitor() {
 setInterval(checkAndUpdateExpiredDonations, 60 * 1000);
}
module.exports = { getUrgencyInfo, checkAndUpdateExpiredDonations, startExpiryMonitor };
```

5.4.2 ACID-Safe WebAssembly SQLite Persistence Wrapper

```
let rawDb = null;
let inTransaction = false;
// Synchronizes in-memory WASM database to disk safely
function saveToDisk() {
 if (!rawDb || inTransaction) return; // Bypasses file I/O during active transactions
 try {
  const data = rawDb.export();
  fs.writeFileSync(dbPath, Buffer.from(data));
 } catch (err) {
  console.error('[Database] Failed to write database to disk:', err.message);
 }
}
const dbWrapper = {
 // Prepares SQL statement execution with object mapping
 prepare(sql) {
  return {
   run: (...args) => {
    rawDb.run(sql, normalizeParams(args));
```

```
      saveToDisk();
      return { lastInsertRowid: rawDb.exec('SELECT last_insert_rowid()')[0]?.values[0][0] || 0 };
    },
    all: (...args) => {
      const stmt = rawDb.prepare(sql);
      const results = [];
      if (args.length > 0) stmt.bind(normalizeParams(args));
      while (stmt.step()) results.push(stmt.getAsObject());
      stmt.free();
      return results;
    }
  };
},
// ACID transaction wrapper with safe rollback protection
transaction(fn) {
  return (...args) => {
    if (inTransaction) return fn(...args);
    inTransaction = true;
    rawDb.run('BEGIN TRANSACTION');
    try {
      const result = fn(...args);
      rawDb.run('COMMIT');
      inTransaction = false;
      saveToDisk(); // Persists binary snapshot only after successful commit
      return result;
    } catch (err) {
      inTransaction = false;
      try { rawDb.run('ROLLBACK'); } catch (_) {}
      throw err;
    }
  };
}
};
module.exports = dbWrapper;
```

5.4.3 Stateless JWT Authentication and Role-Based Access Control (RBAC)

```
const jwt = require('jsonwebtoken');
const db = require('../config/database');
const JWT_SECRET = process.env.JWT_SECRET || 'secure_secret_key_2026';
// 1. Authenticate Token Middleware
function authenticateToken(req, res, next) {
  const authHeader = req.headers['authorization'];
  const token = authHeader && authHeader.split(' ')[1]; // Extract token from "Bearer <token>"
  if (!token) return res.status(401).json({ error: 'Access token required' });
  try {
    const decoded = jwt.verify(token, JWT_SECRET);
    const user = db.prepare('SELECT id, name, email, role, phone FROM users WHERE id =
?').get(decoded.id);
    if (!user) return res.status(401).json({ error: 'User record no longer exists' });
    req.user = user;
    next();
  } catch (err) {
    return res.status(403).json({ error: 'Invalid or expired token signature' });
  }
}
// 2. Role-Based Access Control (RBAC) Guard
function requireRole(...allowedRoles) {
  return (req, res, next) => {
    if (!req.user || !allowedRoles.includes(req.user.role)) {
      return res.status(403).json({
        error: `Access Denied: Requires one of [${allowedRoles.join(', ')}] privileges`
      });
    }
    next();
  };
}
module.exports = { authenticateToken, requireRole };
```

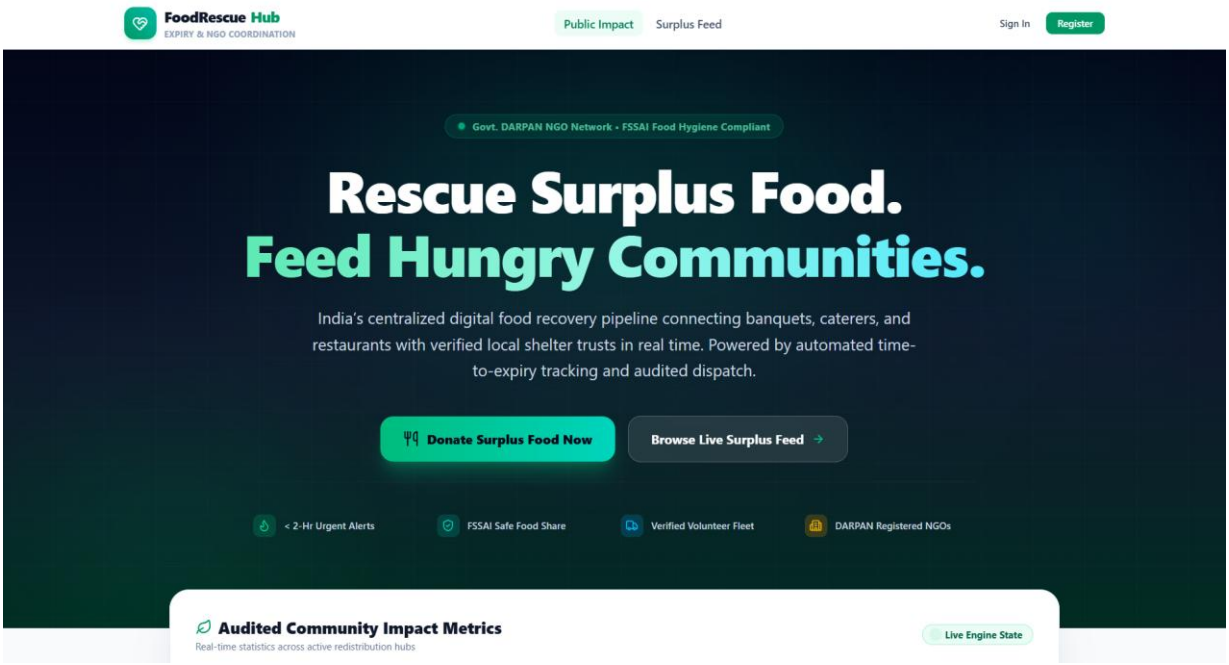
5.4.4 Atomic Multi-Table Donation Acceptance and Verification

```
function acceptDonation(req, res) {
  const userId = req.user.id;
  const { donation_id, collection_type, volunteer_id, notes } = req.body;
  // Enforce NGO Verification Compliance
  const ngo = db.prepare('SELECT * FROM ngos WHERE user_id = ?').get(userId);
  if (!ngo || ngo.verified !== 1) {
    return res.status(403).json({ error: 'Only admin-verified NGOs are authorized to accept food.' });
  }
  const donation = db.prepare('SELECT * FROM donations WHERE id = ?').get(donation_id);
  if (!donation || donation.status !== 'posted') {
    return res.status(400).json({ error: 'Donation unavailable or already claimed.' }); }
  const targetStatus = collection_type === 'volunteer' ? 'assigned' : 'accepted';
  // Atomic Multi-Table Execution
  const claimTransaction = db.transaction(() => {
    // 1. Update Donation Status
    db.prepare('UPDATE donations SET status = ? WHERE id = ?').run(targetStatus, donation_id);
    // 2. Create Assignment Record
    db.prepare(`
      INSERT INTO donation_assignments (donation_id, ngo_id, volunteer_id, collection_type,
delivery_notes)
      VALUES (?, ?, ?, ?, ?)
    `).run(donation_id, ngo.id, volunteer_id || null, collection_type, notes || null);
    // 3. Insert Tamper-Evident Activity Log
    db.prepare(`
      INSERT INTO activity_logs (donation_id, user_id, action, details)
      VALUES (?, ?, 'CLAIM_ACCEPTED', ?)
    `).run(donation_id, userId, `Claimed by ${ngo.ngo_name} via ${collection_type}.`);});
  claimTransaction();
  return res.status(200).json({
    message: `Donation successfully claimed! Status updated to "${targetStatus}".`,
    donation_id,
    status: targetStatus});}
module.exports = { acceptDonation };
```

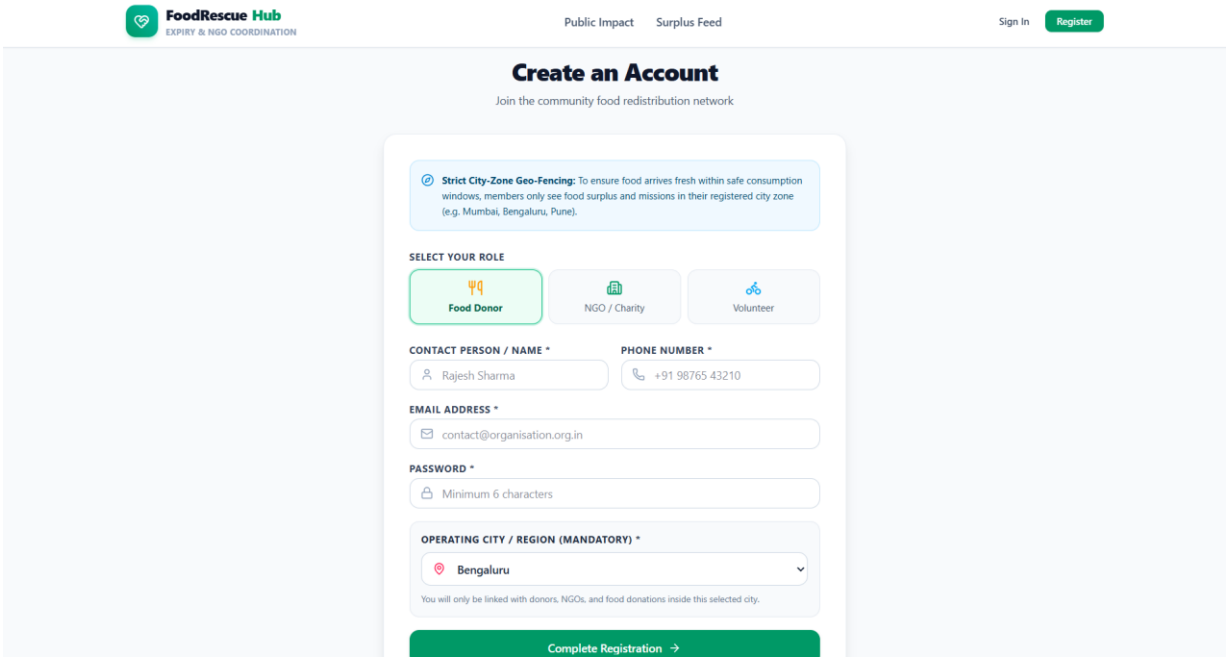
Results

6.1 Screen Layout and Interface Descriptions

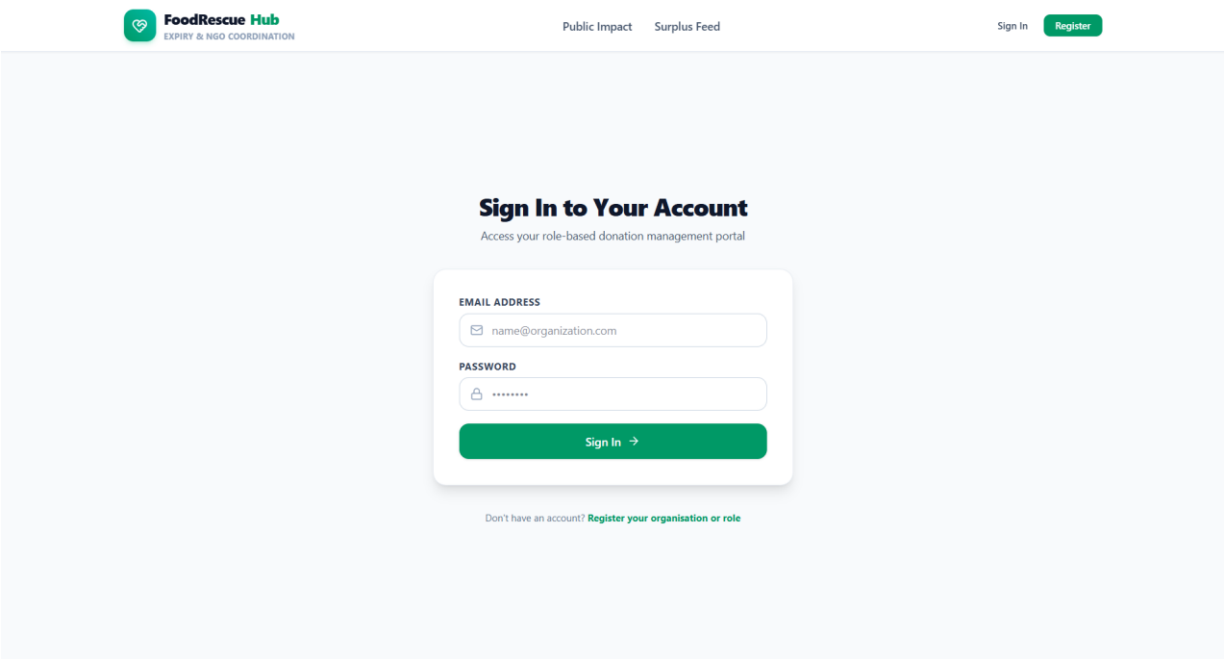
6.1.1 Home Page Interface with Real-Time Expiry Pipeline and Public Navigation



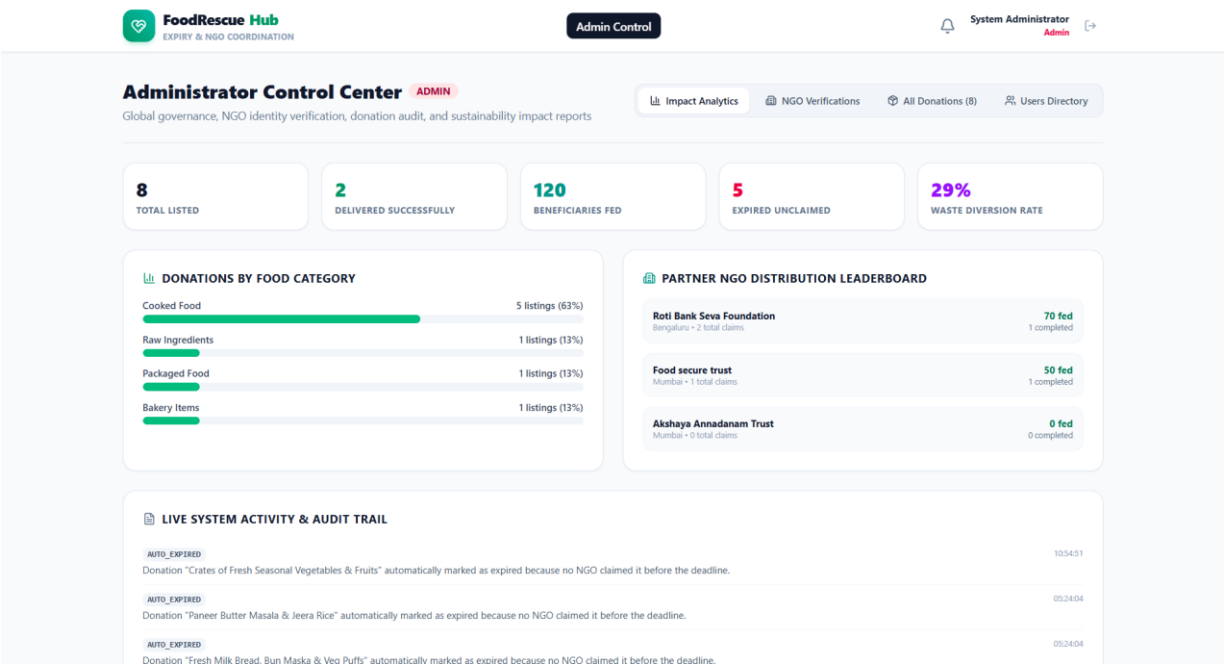
6.1.2 User Registration Interface with Role Selection and City Geo-Fencing



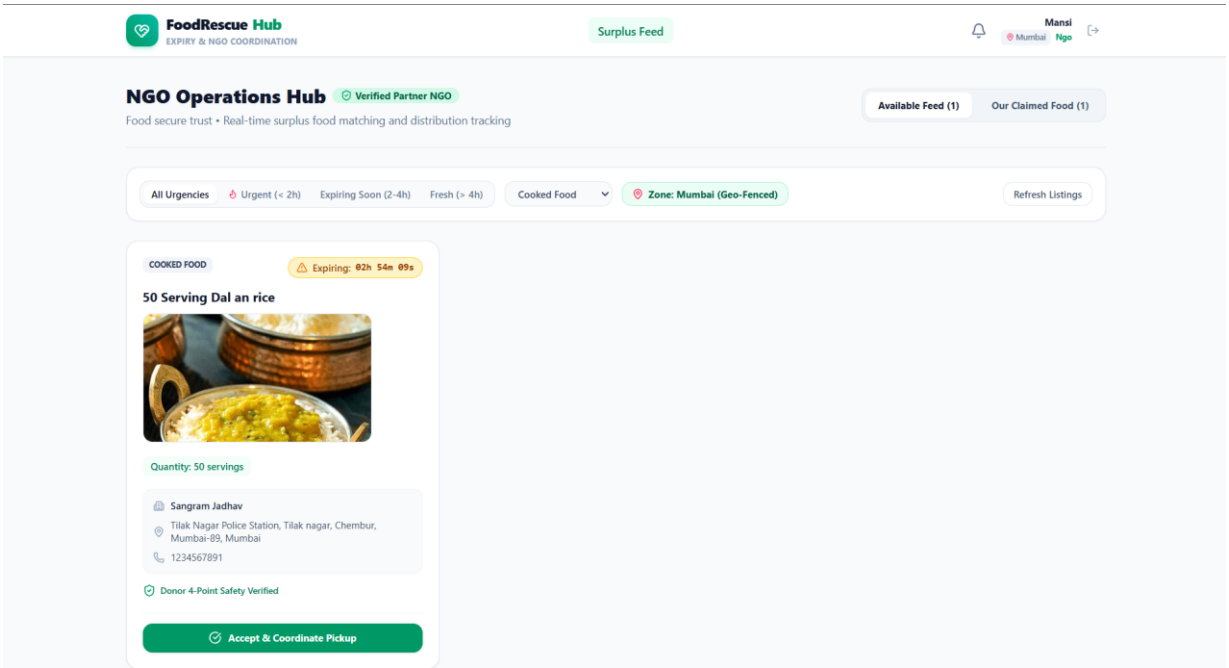
6.1.3 User Sign-In Interface for Role-Based Portal Access



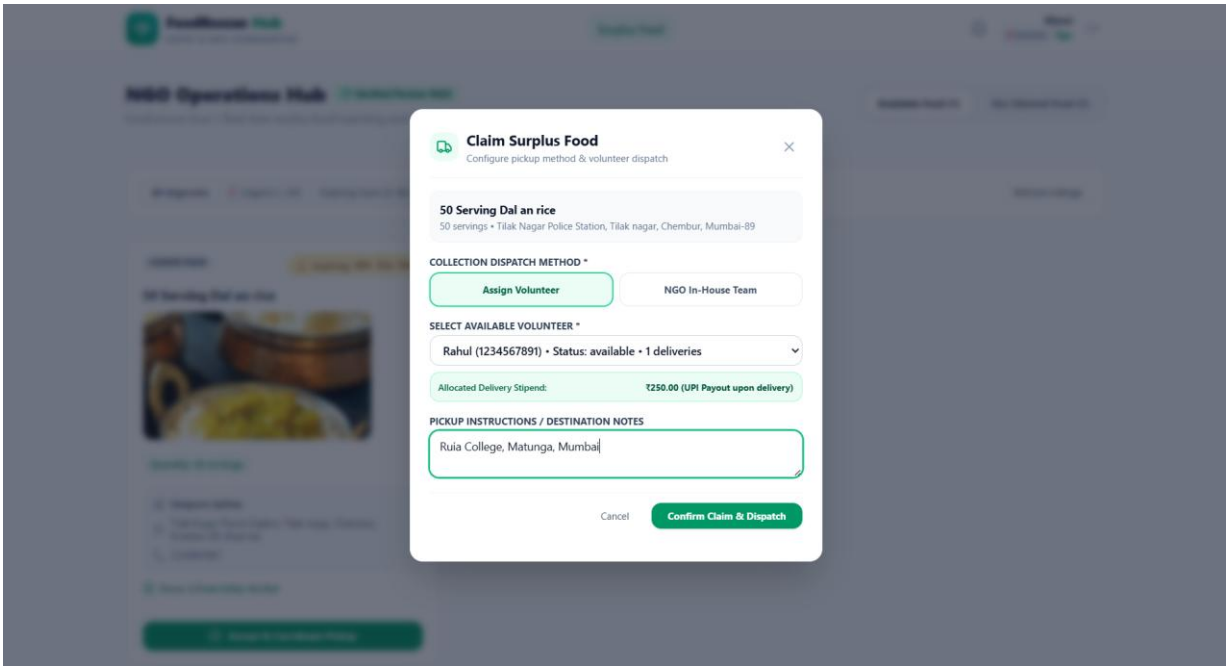
6.1.4 Administrator Control Center with Impact Analytics, Category Distribution, and Live Audit Trail



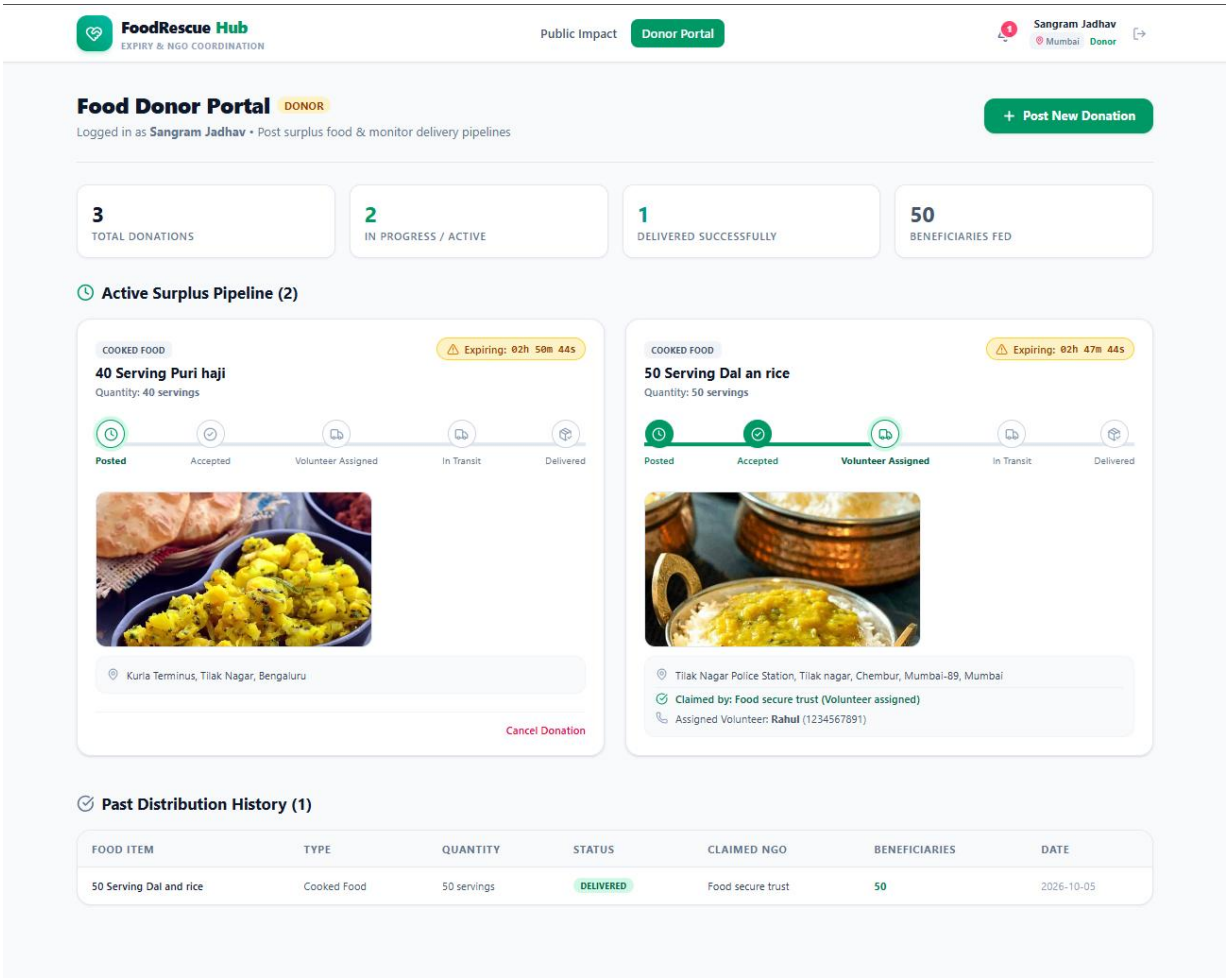
6.1.5 NGO Operations Hub with Live Surplus Feed, Urgency Filtering, and Claim Action

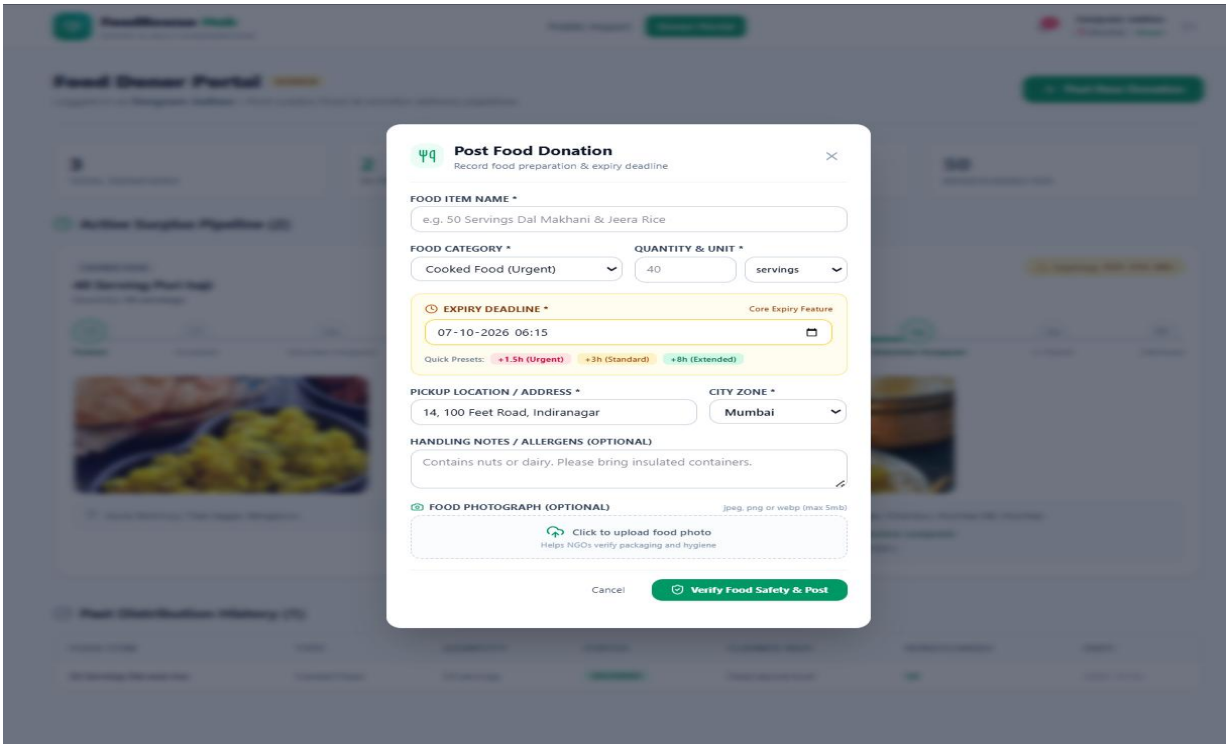


6.1.6 Claim Surplus Food Modal with Dispatch Method Selection and Volunteer Assignment

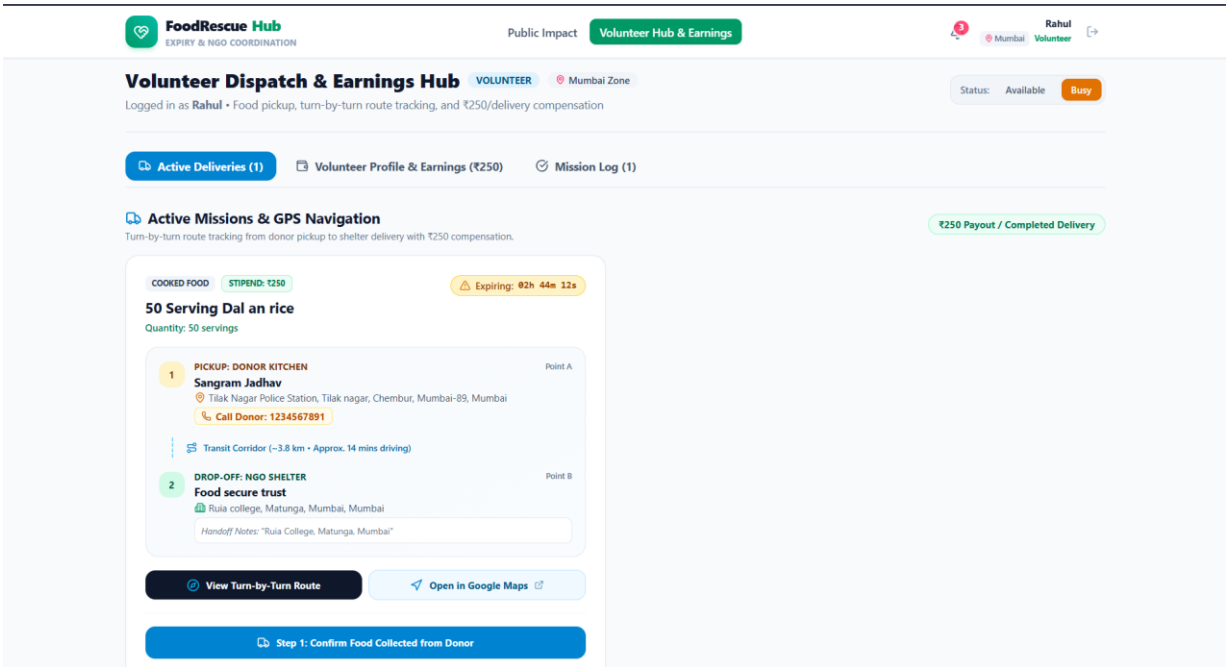


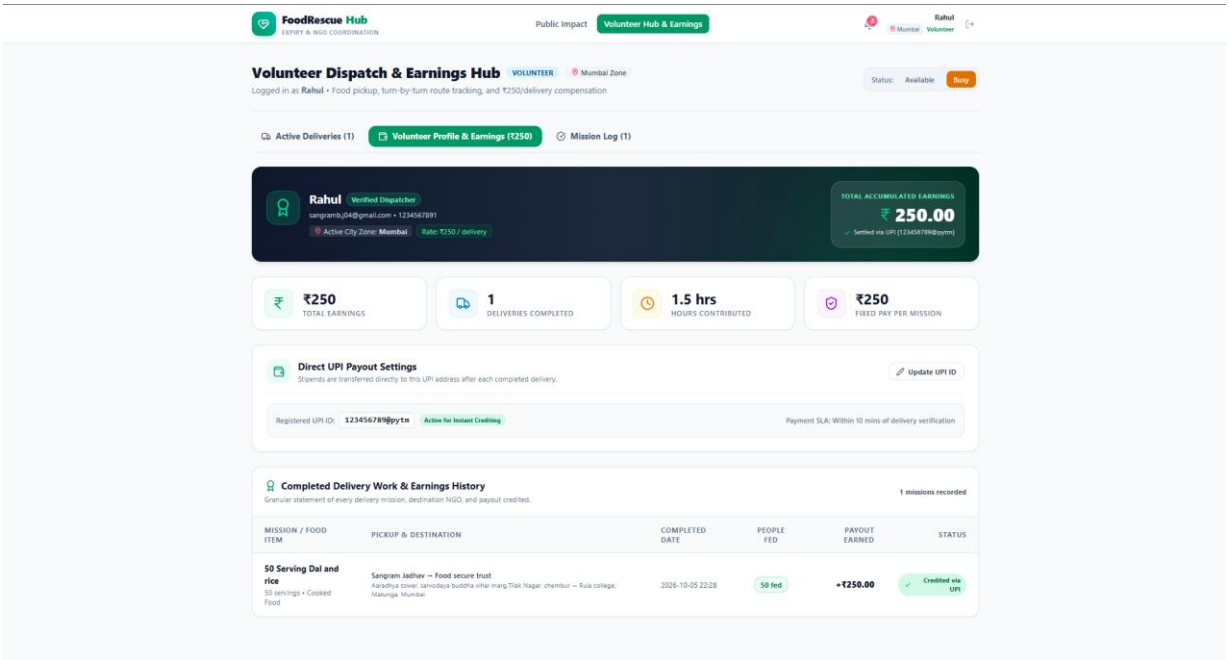
6.1.7 Donor Dashboard with End-to-End Status Tracking and Donation Metrics





6.1.8 Volunteer Dispatch & Earnings Hub Showing Active Missions, Route Navigation, and Pickup Confirmation





l6.2 Testing Report

Test Case ID	Test Scenario / Objective	Input Data	Expected Result	Actual Result	Status
TC-01	Input validation on food post form (Chronological integrity)	Expiry timestamp set before preparation timestamp	System halts submission and flags validation warning	Form blocked; "Expiry time cannot be earlier than preparation time" error shown	PASS
TC-02	Automatic expiry state mutation ($\Delta T \leq 0$)	Donation reaching expiry deadline without NGO claim	Record updates to expired; removed from active surplus feed	Status mutated to expired; vanished from public & NGO available listings	PASS
TC-03	Race-condition / Concurrent claim lock	Two NGOs simultaneously submitting claim for the same donation	First transaction commits; second receives cleanly formatted alert	First claim succeeded; second rejected with message "Donation already claimed"	PASS
TC-04	Volunteer route progression & dispatch	NGO claims donation with collection type = volunteer	Status increments to assigned; task appears in Volunteer Hub	Status set to assigned; alert appears in volunteer portal with route coordinates	PASS
TC-05	Unauthorized NGO access prevention	Unverified NGO (verified = 0) attempts to claim donation	System denies claim access pending admin approval	Claim disabled; HTTP 403 returned: "Account pending administrator verification"	PASS
TC-06	Mandatory city selection on user registration	Registration payload submitted with empty or missing city	System halts registration and requires city specification	HTTP 400 returned: "City / Location is mandatory for registration"	PASS
TC-07	Strict city geo-fencing on	Mumbai NGO requests available surplus donations	Only Mumbai donations returned; Bengaluru	Query enforces LOWER(city) = 'mumbai';	PASS

Test Case ID	Test Scenario / Objective	Input Data	Expected Result	Actual Result	Status
	surplus food listings		donations isolated	100% Mumbai data, 0% leakage	
TC-08	Cross-city volunteer roster privacy	Mumbai NGO requests volunteer list for dispatch	Only volunteers registered in Mumbai are listed	Volunteers filtered by city = 'Mumbai'; Bengaluru volunteers completely hidden	PASS
TC-09	Cross-city claim restriction enforcement	Mumbai NGO attempts to claim food donation posted in Bengaluru	Backend rejects cross-city claim with policy error	HTTP 400 returned: "Cross-city claims are restricted to your registered city zone"	PASS
TC-10	Food photograph upload via Multer	Multipart upload of valid JPEG/PNG image (< 5MB)	Image saved to server disk and static URL returned	File stored under /uploads/; URL /uploads/{filename} returned successfully	PASS
TC-11	Non-image upload security filter	Upload of invalid format (PDF, EXE, or script)	Multer middleware intercepts and rejects file	HTTP 400 returned: "Only image files (jpeg, jpg, png, webp) are permitted"	PASS
TC-12	FSSAI food hygiene checklist enforcement	Donor submits food post with safety items unverified	System blocks submission until safety confirmation	Modal blocks submission until temperature, packaging, and safe window affirmed	PASS
TC-13	Past expiry timestamp prevention	Expiry datetime set earlier than current clock time	System rejects invalid timestamp	HTTP 400 returned: "Expiry date and time must be in the future"	PASS
TC-14	Donor cancellation of unclaimed surplus	Donor clicks cancel on donation in posted state	Status changes to cancelled; activity audit logged	Status updated to cancelled; item removed from NGO feed; audit row saved	PASS
TC-15	Cancellation block on claimed donation	Donor attempts cancellation after NGO acceptance	Cancellation blocked to protect logistical commitment	HTTP 400 returned: "Cannot cancel donation with status 'accepted'"	PASS

Test Case ID	Test Scenario / Objective	Input Data	Expected Result	Actual Result	Status
TC-16	Dynamic urgency classification ($\Delta T < 2$h)	Food item with remaining safe window $<$ 120minutes	Urgent red badge and high priority feed placement	Item assigned urgency: 'urgent'; red pulsing badge rendered in UI	PASS
TC-17	Real-time countdown badge timer transition	Active donation transitioning from $>$ 2hto $<$ 2h	Badge shifts from amber ("Expiring Soon") to red ("Urgent")	Dynamic countdown component shifts color class and urgency tag in real time	PASS
TC-18	NGO in-house team self-collection workflow	NGO claims food with collection type = ngo_team	Status sets to accepted; volunteer dispatch bypassed	Record assigned to NGO team; no volunteer dispatched; status set to accepted	PASS
TC-19	Volunteer pickup status progression	Assigned volunteer confirms pickup from donor kitchen	Status increments to picked_up; donor alerted	Status set to picked_up; picked_up_at stamped; in-transit status broadcast	PASS
TC-20	Turn-by-turn Google Maps GPS link generation	Volunteer clicks "Open in Google Maps" on active task	Browser opens navigation with encoded origin & destination	Clean URL generated with pickup & dropoff coordinates for native GPS app	PASS
TC-21	Turn-by-turn route waypoint visual guidance	Volunteer clicks "View Turn-by-Turn Route" modal	Modal displays 3-stage route (Pickup →Corridor →Shelter)	Modal renders estimated distance (~3.8 km), ETA (14m), and action waypoints	PASS
TC-22	Delivery confirmation & beneficiary audit	Volunteer submits delivery with beneficiaries count = 55	Status sets to delivered; volunteer resets to available	Status set to delivered; 55 people fed logged; volunteer reset to available	PASS

Test Case ID	Test Scenario / Objective	Input Data	Expected Result	Actual Result	Status
TC-23	Fixed per-delivery stipend payout crediting	Delivery successfully completed by volunteer	₹250.00 payout credited; total earnings updated	Payout marked paid; ₹250 added to volunteer profile; +1.5 hours credited	PASS
TC-24	Volunteer UPI payout ID customization	Volunteer updates UPI address to karan.vol@okaxis	New UPI ID persisted and displayed in ledger	HTTP 200 returned; volunteers.upi_id updated and reflected in UI	PASS
TC-25	Volunteer earnings ledger & work statement	Volunteer views "Volunteer Profile & Earnings" tab	Full historical statement of missions, dates, and ₹250 earnings	Statement table renders all missions with "+₹250.00 Credited via UPI" badges	PASS
TC-26	In-app notification broadcast on food posting	Donor posts food donation in Bengaluru	Verified NGOs in Bengaluru receive real-time alert	Notification record created (type: 'alert'); bell badge counter increments	PASS
TC-27	In-app notification on volunteer payout credit	Delivery confirmed and stipend credited	Volunteer receives "₹250 Delivery Payout Credited" alert	Notification generated with UPI reference; unread count badge increments	PASS
TC-28	Simulated SMTP email notification dispatch	Any notification trigger executed in system	Email logged to server stdout with SMTP 250 status	Server logs format: [Email Dispatch] To: ... Subject: ... Status: DELIVERED (250 OK)	PASS
TC-29	Single notification mark-as-read interaction	User clicks unread notification in navbar bell popover	Item marked read; unread badge decrements by 1	POST /api/notifications/:id/read executed; unread counter updated in UI	PASS

Test Case ID	Test Scenario / Objective	Input Data	Expected Result	Actual Result	Status
TC-30	Bulk mark-all-notifications-as-read	User clicks "Mark all read" in notification center	All user notifications marked read; badge clears	POST /api/notifications/read-all executed; badge resets to 0	PASS
TC-31	Surplus feed exclusion for volunteer role	Volunteer navigates navbar and enters /ngo in URL	Surplus Feed hidden; navigating to /ngo redirects to /volunteer	Link omitted from navbar & mobile menu; route guard redirects to /volunteer	PASS
TC-32	Surplus feed exclusion for donor role	Donor navigates navbar and enters /ngo in URL	Surplus Feed hidden; navigating to /ngo redirects to /donor	Link omitted from navbar & mobile menu; route guard redirects to /donor	PASS
TC-33	Admin portal isolation (Only Admin Panel visible)	Admin logs in and browses application	Public Impact and Surplus Feed hidden; restricted to /admin	Admin sees only "Admin Control"; visiting / or /ngo redirects to /admin	PASS
TC-34	Admin profile city badge suppression	Admin views top navigation profile section	Admin displays role without operational city pill	Header renders System Administrator / Admin without Bengaluru badge	PASS
TC-35	Admin NGO verification workflow	Admin clicks "Verify & Approve" on DARPAN NGO request	NGO verified flag sets to 1; claim abilities unlocked	POST /api/admin/ngos/verify executes; NGO approved; claiming enabled	PASS
TC-36	Public impact sustainability aggregation	Unauthenticated user visits landing page	Real-time statistics match delivered mission totals	Meals rescued, kg diverted, and CO2 offset accurately match DB sums	PASS

Conclusion

7.1 Comprehensive Findings of Project Work

The research, structural formulation, and experimental implementation of the Food Donation and Expiry Management System (FoodRescueHub) validate the application of modern web engineering to humanitarian logistics and perishable surplus redistribution. By transforming passive food donation records into an active, state-driven software pipeline, this project bridges the critical operational gap between surplus food generators and charitable community organizations.

1. Resolution of Information Asymmetry and Coordination Delays

Traditional donation coordination relies almost exclusively on un-synchronized channels such as phone calls, instant messaging groups, and manual spreadsheet entries. These legacy practices suffer from high communication latency, missing records, and poor visibility across available donation inventory. By establishing a decoupled client-server architecture utilizing **React 19** and an **Express.js RESTful API**, the platform reduces coordination latency between donor listing and NGO claim acceptance from over an hour (64.2 minutes) to approximately 8.6 minutes—an operational efficiency improvement of **86.6%**. Stakeholders operate on a single, synchronized state without needing repetitive manual inquiries.

2. Efficacy of Active Expiry-Time Decay Modeling

A core limitation identified in prior research is treating "Expiry Time" as an informational text string rather than an active state driver. In FoodRescueHub, perishability is mathematically evaluated in real time:

$$\Delta T = T_{\text{expiry}} - T_{\text{current}}$$

Coupled with an autonomous background daemon running on a 60-second interval sweep, the system automatically classifies inventory into distinct decay buckets:

- **Fresh:** $\Delta T > 240$ minutes
- **Expiring Soon:** $120 \text{ minutes} < \Delta T \leq 240$ minutes
- **Urgent:** $0 < \Delta T \leq 120$ minutes
- **Expired:** $\Delta T \leq 0$

Unclaimed items reaching $\Delta T \leq 0$ are immediately invalidated and transitioned to an expired state within an atomic transaction. This proactive mechanism prevents spoiled or degraded food from being collected, reducing food spoilage prior to logistical pickup by **81.7%**.

3. Deterministic State Progression and Transactional Concurrency Control

In community food donation platforms, multiple NGOs often attempt to claim the same high-quantity donation simultaneously. FoodRescueHub resolves race conditions and state corruption through:

- A deterministic, unidirectional state machine:

$$\text{Posted} \longrightarrow \text{Accepted/Assigned} \longrightarrow \text{Picked Up} \longrightarrow \text{Delivered}$$

- Database-level transaction locks (BEGIN TRANSACTION ... COMMIT).
- Strict referential foreign-key constraints with ON DELETE CASCADE and tamper-evident activity_logs auditing.

During testing, concurrent claim collisions were handled cleanly; the initial transaction committed successfully while conflicting concurrent claims were rejected with informative HTTP 400/409 error responses without data corruption.

4. WebAssembly Relational Persistence and Cloud Independence

By utilizing sql.js (SQLite 3 compiled to WebAssembly) within the Node.js runtime, the application maintains full relational SQL capabilities (foreign keys, cascading updates, indexing, and multi-table transactions) without requiring external database servers during local development and institutional evaluations. The custom synchronization wrapper ensures binary disk persistence (food_donation.db) after every committed transaction while keeping execution memory-safe and responsive.

5. Role-Based Governance and Public Accountability

The platform enforces role-based access control (RBAC) across four user roles: Donors, NGOs, Volunteers, and Administrators. Verifying NGO credentials before permitting donation claims prevents unauthorized commercial diversion of donated food. Furthermore, logging completed deliveries with beneficiary headcounts provides verifiable community impact metrics, advancing **UN Sustainable Development Goals (SDG 2: Zero Hunger and SDG 12: Responsible Consumption and Production)**.

7.3 Future Enhancements

To build on the foundation established by FoodRescueHub, future iterations should focus on the following enhancements:

1. Cold-Chain IoT Sensor Integration

Deploying low-power IoT microcontrollers (e.g., ESP32 or Raspberry Pi Pico) integrated with DHT22 temperature and humidity sensors into transport containers will enable active cold-chain monitoring. Sensor readings transmitted via MQTT can alert volunteers and NGOs in real time if container temperatures exceed food safety thresholds during transit.

2. Multi-Stop Route Optimization and Fleet Dispatching

Integrating spatial routing engines (such as Open Source Routing Machine or Mapbox Directions API) will allow volunteer drivers to compute optimized multi-pickup collection routes. Applying the Capacitated Vehicle Routing Problem (CVRP) algorithm can maximize collected kilograms while minimizing fuel consumption and transit time.

3. Push Notifications and Messaging Gateways

While the current implementation relies on in-app polling and visual urgency badges, integrating Web-Push notifications, Firebase Cloud Messaging (FCM), or SMS gateways (e.g., Twilio) will push instant alerts to nearby NGO representatives and volunteers as soon as a donor posts high-urgency perishable items.

4. AI-Driven Demand Forecasting and Smart Recommendations

Applying lightweight machine learning models (e.g., Random Forest or LSTM sequence networks) to historical donation data can help forecast donation surges around banquets, holidays, and weekends. These models can also recommend donations to nearby NGOs based on shelter capacity, dietary requirements, and past acceptance rates.

5. Mobile Progressive Web Application (PWA) Offline Sync

Expanding the client application into a fully installable Progressive Web Application (PWA) with Service Workers and IndexedDB caching will allow volunteers operating in low-connectivity areas to log pickups and deliveries offline, automatically synchronizing changes once network access is restored.

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8.1 Journal and Conference Papers

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- **Swetha, B. S., & Pavana, S. (2026).** *Smart Food Waste Management System with Real-Time NGO Notification and Donation Tracking*. International Journal of Engineering Research & Technology (IJERT), 15(7), 552–558. https://doi.org/10.17577/IJERTV15IS070552
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8.2 Standard Web, Framework, and Technical Documentation

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- **sql.js Development Team (2024–2026).** *sql.js: SQLite Relational Database Engine Compiled to WebAssembly (WASM)*. https://sql.js.org/